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Education:

North Carolina State University	Zoology with Statistics and Modeling minor	Ph.D.	1994
University of Rhode Island	Oceanography, biological option	M.S.	1990
Trinity College (CT)	Biology	B.S.	1985

Dissertation Title: Causes and consequences of growth variability in young fishes.

Employment:

2025-now	Lead data scientist at WestBook DataViz, LLC
1995-2025	USGS, Ecology section leader, Conte Fish Research Laboratory, Turners, Falls, MA, and adjunct professor Department of Natural Resources Conservation, University of Massachusetts - Amherst
1994-1995	Oak Ridge Postdoctoral Fellow (with Dr. David Conover) State University of New York, Stony Brook, NY
1992-1994	Electric Power Research Institute Fellow North Carolina State University, Raleigh, NC
1990-1992	Wisconsin Sea Grant Fellow North Carolina State University, Raleigh, NC
1987-1990	URI-USEPA Cooperative Research Fellow University of Rhode Island, Kingston, RI
1986-1987	Research Assistant, Center for Coastal Studies Provincetown, MA.

1985-1986 Research Assistant, Dept. Psychology, University of Massachusetts, Amherst, MA.

Professional Activities:

USGS Eastern Region AI/ML Capability Team co-lead
Editor, peerJ, (2013 - now)
Associate Editor, Transactions of the American Fisheries Society (1998 - 2001)
Co-Book review editor, American Fisheries Society (2005 - 2009)

Professional societies:

American Association for the Advancement of Science
Ecological Society of America
American Fisheries Society
AFS Early Life History Section (North East representative)
AFS Southern New England Chapter-At large representative
NASCO Assessment Committee representative
Science advisor to The Nature Conservancy's Connecticut River program

Teaching experience:

Fisheries ecology Teaching Assistant, North Carolina State University, 1992
Individual-based modeling, SUNY Stony Brook, 1995
Ecology of individuals, University of Massachusetts, 1997
Evolution of migration, University of Massachusetts, 1999
Evolution and ecology of migration, University of Massachusetts, 2000
Ecology of diadromous fishes (team taught), University of Massachusetts, 2002, 2004, 2008
Analysis of mark-recapture data, University of Massachusetts, 2004, 2005, 2008
Data management and manipulation (supervised PhD student), University of Massachusetts, 2005
Hierarchical modeling workgroup, University of Massachusetts, 2009
Organismic and Evolutionary Biology Core Ecology class (team taught), UMass, 2009, 2011, 2013, 2015, 2017
Advanced Statistics 3 – Bayesian linear models, UMass, 2010

Web applications:

<https://www.westbrookdataviz.org>

<https://www.usgs.gov/apps/ecosheds>, Spatial hydro-ecological decision system: a modeling system and collection of interactive web applications including:

1. Flow photo explorer
2. Stream temperature data system for NE USA
 - a. Stream temperature model

- b. Stream fish occupancy model
- c. Interactive catchment explorer
- 3. Stream temperature database for AK
- 4. Tagged animal movement explorer
- 5. RESTORE visualization tool
- 6. Native trout vulnerability tool
- 7. Stream crossings explorer
- 8. Fish tagging data visualizations
- 9. Time series explorer

Public Observable notebooks, <https://observablehq.com/@bletcher>

Publications:

- Letcher, Benjamin H., Keith H. Nislow, Matthew J O'Donnell, Michael J Hayden, and Todd L Dubreuil. 2025. Negative Growth in Body Mass of Trout and Salmon in a Small Stream Network. *Canadian Journal of Fisheries and Aquatic Sciences*. <https://doi.org/10.1139/cifas-2024-0376>.
- O'Donnell, M. J., A. M. Regish, S. D. McCormick, and B. H. Letcher. 2025. "How Quickly Do Brook Trout Lose Long-Term Thermal Acclimation?" *Journal of Thermal Biology* 129: 104103. <https://doi.org/10.1016/j.jtherbio.2025.104103>.
- O'Donnell, Matthew J., Amy M. Regish, Stephen D. McCormick, and Benjamin H. Letcher. 2024. Thermal Transfer Rate Is Slower in Bigger Fish: How Does Body Size Affect Response Time of Small, Implantable Temperature Recording Tags? *Ecology of Freshwater Fish* 33, no. 4: e12794. <https://doi.org/10.1111/eff.12794>.
- Childress, Evan S., E. David Demarest, John E. B. Wofford, Nathaniel P. Hitt, and Benjamin H. Letcher. 2024. Strong Variation in Brook Trout Trends across Geology, Elevation, and Stream Size in Shenandoah National Park. *Transactions of the American Fisheries Society*. <https://doi.org/10.1002/tafs.10460>.
- Zhao, Xiaohu, Kebin Jia, Benjamin Letcher, Jennifer Fair, and Xiaowei Jia. 2024. Bringing Vision to Climate: A Hierarchical Model for Water Depth Monitoring in Headwater Streams. *Information Fusion*, 102448. <https://doi.org/10.1016/j.inffus.2024.102448>.
- Lonthair, Joshua K., Nicholas C. Wegner, Brian S. Cheng, Nann A. Fangue, Matthew J. O'Donnell, Amy M. Regish, John D. Swenson, et al. 2024. The Gill Oxygen Limitation Hypothesis Remains Unsupported by Experimental Evidence. Response to 'Testing Mechanistic Theories Must Be Based on Correct Interpretations. *Journal of Experimental Biology* 227, 16, jeb248173. <https://doi.org/10.1242/jeb.248173>.
- Lonthair, Joshua K., Nicholas C. Wegner, Brian S. Cheng, Nann A. Fangue, Matthew J. O'Donnell, Amy M. Regish, John D. Swenson, et al. 2024. Smaller Body Size under Warming Is Not Due to Gill-

- Oxygen Limitation in a Cold-Water Salmonid. *Journal of Experimental Biology* 227(4): jeb246477. <https://doi.org/10.1242/jeb.246477>.
- Shi, Weili, Zhongliang Zhou, Benjamin H. Letcher, Nathaniel Hitt, Yoichiro Kanno, Ryo Futamura, Osamu Kishida, Kentaro Morita, and Sheng Li. 2024. Aging Contrast: A Contrastive Learning Framework for Fish Re-Identification Across Seasons and Years. In *AI 2023: Advances in Artificial Intelligence*, edited by Tongliang Liu, Geoff Webb, Lin Yue, and Dadong Wang, 252–64. Singapore: Springer Nature, 2024. https://doi.org/10.1007/978-981-99-8388-9_21.
- Valentine, George P., Xinyi Lu, Evan S. Childress, C. Andrew Dolloff, Nathaniel P. Hitt, Matthew A. Kulp, Benjamin H. Letcher, et al. 2024. Spatial Asynchrony and Cross-Scale Climate Interactions in Populations of a Coldwater Stream Fish. *Global Change Biology* 30(1): e17029. <https://doi.org/10.1111/gcb.17029>.
- Letcher, Benjamin H., Keith H. Nislow, Matthew J. O'Donnell, Andrew R. Whiteley, Jason A. Coombs, Todd L. Dubreuil, and Daniel B. Turek. Identifying Mechanisms Underlying Individual Body Size Increases in a Changing, Highly Seasonal Environment: The Growing Trout of West Brook. *Journal of Animal Ecology* 92, no. 1 (2023): 78–96. <https://doi.org/10.1111/1365-2656.13833>.
- Mejia, F.H., Ouellet, V. et al. 2023. Closing the Gap between Science and Management of Cold-Water Refuges in Rivers and Streams. *Global Change Biology*. <https://doi.org/10.1111/gcb.16844>.
- Mamoozadeh, Nadya R., Andrew R. Whiteley, Benjamin H. Letcher, David C. Kazyak, Charlene Tarsa, and Mariah H. Meek. 2023. A New Genomic Resource to Enable Standardized Surveys of SNPs across the Native Range of Brook Trout (*Salvelinus Fontinalis*). *Molecular Ecology Resources*, 1755-0998.13853. <https://doi.org/10.1111/1755-0998.13853>.
- He, Erhu, Yue Wan, Benjamin H. Letcher, Jennifer H. Fair, Yiqun Xie, and Xiaowei Jia. “CGS: Coupled Growth and Survival Model with Cohort Fairness,” 6:5986–94, 2023. <https://doi.org/10.24963/ijcai.2023/664>.
- Gupta, A, Chang, T., Walker, J. D., and Letcher, B.H., 2022. Towards Continuous Streamflow Monitoring with Time-Lapse Cameras and Deep Learning, ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS), <https://doi.org/10.1145/3530190.3534805>
- Zhou, Zhongliang, Nathaniel P. Hitt, Benjamin H. Letcher, Weili Shi, and Sheng Li. Pigmentation-Based Visual Learning for *Salvelinus Fontinalis* Individual Re-Identification. In *2022 IEEE International Conference on Big Data (Big Data)*, 6850–52, 2022. doi.org/10.1109/BigData55660.2022.10020966.
- Zhao, Xiaohu, Kebin Jia, Benjamin Letcher, Jennifer Fair, Yiqun Xie, and Xiaowei Jia. “VIMTS: Variational-Based Imputation for Multi-Modal Time Series.” In *2022 IEEE International Conference on Big Data (Big Data)*, 349–58, 2022. <https://doi.org/10.1109/BigData55660.2022.10020834>.

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- Letcher et al. 2022, Identifying mechanisms underlying individual body size increases in a changing, highly-seasonal environment: the growing trout of West Brook, *Journal of Animal Ecology*, 92(1), 78-96, <https://doi.org/10.1111/1365-2656.13833>
- Letcher et al., 2022, Cohort strength and body size in co-occurring salmonids in a small stream network: variation in space and time, *CJFAS*, 79(1), <https://doi.org/10.1139/cjfas-2020-0418>.
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Book chapters:

- Forseth, T., B.H. Letcher and M. Johansen, 2010. The behavioural flexibility of salmon growth, in *Atlantic Salmon Ecology*, O. Aas, S. Einum, A. Klemetsen, and J. Skurdal, Eds., Wiley-Blackwell, Oxford, UK.
- Kovach, R., et al. 2019. Climate change and the future of trout and char, Pages 685-716 in J.L. Kershner, J.E. Williams, R.E. Gresswell, and J Lobon-Cervia, editors. *Trout and char or the world*, American fisheries society, Bethesda, MD.

Presentations:

Letcher, B.H. et al. Overview of brook trout research and AI's role in advancing aquatic science, **invited**, Trout Unlimited brook trout workshop, Charlemont, MA, August 2024

Letcher, B.H. et al. AI's role in advancing aquatic science, **invited symposium**, Ecological Society of America annual meeting, Aug 2024.

Letcher, B.H. et al. Estimating stream flow using time-lapse imagery and machine learning with the 'Flow Photos Explorer', **invited**, NEAFWA annual meeting, Hyannis, MA, April 2024.

Letcher, B.H. and J.D. Walker, Data systems and Alaska stream temperature, **invited**, AK CASC project meeting, Anchorage AK, Feb 2024.

Letcher, B.H. and J.D. Walker, TAME - Tagged Animal Movement Explorer, **Invited**, Smithsonian Movement for Life Quarterly meeting, Dec 2023

Letcher, B.H. et al. Eyes on streams: using low-cost time-lapse imagery to estimate streamflow in headwaters, **Invited**, USFS Northeast region Seminar Series, Oct 2023.

Letcher, B.H. et al. Eyes on streams: using low-cost time-lapse imagery to estimate streamflow in headwaters, **Invited**, USGS National Innovation Center Seminar Series, June 2023.

Letcher, B.H., Tagging studies to understand fish growth and survival, **Invited**, Trout Unlimited Squant-a-Tissit Chapter Meeting, May 2023.

Letcher, B.H. et al., Do we need tributary-specific flow estimates to understand trout demography? Biannual meeting of "Advances in the population ecology of stream salmonids", Mallorca, Spain, May 2023.

Letcher, B.H. et al. Coldwater climate refugia, **Invited**, USGS-EPA Northeast region joint conference, Boston, MA. April 2023.

Letcher, B.H. et al. Machine learning for individual fish recognition, **Invited**, USGS National Innovation Center Seminar Series, Dec 2022.

Letcher, B.H. et al., Empowering anglers and assessing fish population trends and conservation needs using AI to identify individual fish, **Invited**, Northeast Region Science, mini-series, Nov 2022

Letcher, B.H. et al., Low-cost Streamflow Monitoring with Deep Learning Computer Vision Modeling, **Invited**, Regional EPA meeting on ORD research, Oct 2022.

Letcher, B.H. et al. Towards joint flow/temperature models for the northeast, **Invited**, Eastern Brook Trout Joint Venture Meeting, April 2022.

Letcher, B.H. et al. Estimating stream flow from images in headwater streams. Atlantic salmon ecosystems forum, Jan 2022.

Letcher, B.H., Coldwater climate refugia in stream networks, **Invited**, Trout Unlimited forum on USGS science, Dec 2021.

Letcher, B.H., Modeling, understanding, and communicating individual tagging data: why are our fish getting bigger? **Invited**, NOAA Seminar Series, Santa Cruz, CA, December 2021.

Letcher, B.H., Overview of SHEDS data visualization and interactive tools: data systems for better decisions, **Invited**, USGS Tribal Engagement Webinar, October, 2021.

Letcher, B.H., Should/can USGS be a leader in science communication and interactive data visualization? **Invited**, Colorado River Basin Technology Collaboration Meeting, July, 2021.

Letcher, B.H. Small streams are important but understudied: a low-cost image-based stream gaging system for headwaters, **Invited Plenary**, Earth Science Information Partners annual meeting, July, 2021. <https://youtu.be/hpxTLNjS0Gk?t=1612>

Letcher, B.H. et al., Economic Valuation of Ecosystem Services in the Delaware River Basin, **Invited**, USGS CASC fish group meeting, July, 2021.

Letcher, B.H. et al., Estimating stream flow from images using Deep Learning: progress to date, **Invited**, Environmental Law Institute Annual Meeting, June, 2021.

Letcher, B.H., The fundamentals of design for scientific data visualization, co-Lead, USGS Community for Data Integration Annual Workshop, May, 2021.

Letcher, B.H., A data system for AI_ML models of stream flow based on images, USGS Community for Data Integration Annual Workshop, May, 2021.

Letcher, B.H., ecoSheds: a data/modelling/visualization system for understanding stream temperature, etc., **Invited**, USGS Chesapeake Bay data meeting, August, 2020.

Letcher, B.H., Should/can USGS be a leader in science communication and interactive data visualization? **Invited**, Joint Science Center meeting, March, 2020.

Letcher, B.H., Cross-filters! Use in interactive data visualizations, **Invited**, Pacific Northwest Aquatic Monitoring Partnership data visualization webinar series, Nov 2019.

Letcher, B.H., An interactive mapping tool: overview of the interactive catchment explorer, **Invited**, USGS Community for Data Integration webinar series, Nov 2019.

Letcher, B.H., Data visualization: understanding and communicating complex data and models, **Invited**, Leetown Science Center Webinar series, June 2019.

Letcher, B.H. et al., The growing fish of the West Brook: what influences body size in a small stream network, **Invited**, Population ecology of stream salmonids V, Granada, Spain, May 2019

Letcher, B.H., Novel Approaches for Estimating and Visualizing Climate Vulnerability for Aquatic Systems, **Invited**, NE Climate Adaptation Science Center, Amherst MA, Nov 2018.

Letcher, B.H., Flow pictures explorer – can we use computer vision to estimate stream flow? **Invited**, Webinar, Regional Monitoring Network, May 2018.

Letcher, B.H., What limits Brook Trout in their native range? **Invited**, Departmental Seminar, Utah State University, Logan, Utah, April 2018.

Letcher, B.H. et al., Generalizability of stream flow effects: how strong is small-scale variation? **Invited**, Society of Freshwater Science annual meeting, Raleigh, NC, June 2017.

Letcher, B.H., Aquatic connectivity. **Invited**, USGS Northeast Climate Science Center 5-year Science meeting, Amherst, MA, May 2017.

Letcher, B.H. et al., What does small-scale variation in demographic response to environmental drivers mean in a changing climate? **Invited**, American Water Resources Association specialty conference on Aquatic Connectivity, Snowbird, UT, April, 2017

Letcher, B.H., Regional databases and models: leveraging information, or 'Why should I contribute to a regional database'?, **Invited plenary**, New England Association of Environmental Biologists, Hartford, CT, March 2017.

Letcher, B.H., Decision support tools in ecosheds.org, **Invited**, Connecticut Conference on Natural Resources, Storrs, CT, March 2017.

Letcher, B.H. and J. D. Walker, Data visualizations of complex data: useful or just cute? **Invited**, Biennial Connecticut River Research Forum, Hadley, MA, March 2017.

Letcher, B.H., Visualizing complex environmental data, **Invited**. NY DEC annual regional managers meeting, New Paltz NY, May 2016.

Letcher, B.H., Evolutionary ecology of brook trout: what can individuals teach us about population persistence, **Invited**, Smith College Biology Department Seminar Series, March 2016

Letcher, B.H., Visualizing data and model results for improved understanding, **Invited**, Annual meeting of the Eastern Brook Trout Joint Venture, Hadley, MA, Sept., 2015.

Letcher, B.H. et al., Developing robust estimates of population response to environmental change: moving towards integrating across scales and data types, **Invited**, American Fisheries Society Annual meeting, Portland, OR, August, 2015.

Letcher, B.H. et al., Developing robust estimates of population response to environmental change: moving towards integrating across scales and data types, Advances in population ecology of stream salmonids, Girona, Spain, May, 2015.

Letcher, B.H. et al. Multi-scale approaches to understanding environmental effects on native brook trout population persistence, **Invited**, Atlantic salmon Ecosystem research forum, Northboro, MA, Mar 2015.

Letcher, B.H., Developing robust estimates of population response to environmental change: moving towards integrating across scales and data types, University of Massachusetts, Department of Math and Statistics, **Invited**, Amherst MA, March, 2015.

Letcher, B.H. Why do large fish need small streams: spatial demography of native brook trout, Colorado State University, Department of Fish, Wildlife and conservation Biology, **Invited**, Fort Collins, CO, Sept. 2014

Letcher, B.H. et al. Stream flow and temperature effects on salmonid population dynamics: integrated modeling across scales and data types, Annual American Fisheries Society Meeting, Quebec City, Quebec, Aug, 2014.

Letcher, B.H. et al. A regional data/modeling system for identifying climate change resilient streams, Connecticut Watershed Volunteer Annual meeting, **Invited**, Hartford, CT, July 2014.

Letcher, B.H. et al., Stream temperature modeling and brook trout response, Joint stream temperature EPA/USFWS/USGS meeting, **Invited**, Hadley MA, April 2014.

Letcher, B.H. et al. Brook trout population dynamics: Integrated modeling across scales and data types, **Invited**, North East Association of Fisheries and Wildlife Agencies, Portland, ME, April, 2014

Letcher, B.H. et al. Environmental drivers of brook trout dynamics, modeling across scales, **Invited**, Adirondack Research Forum, Old Forge, NY, February, 2014

Letcher, B.H. et al., Stream temperature breakpoint modeling, Maine temperature modeling workshop, Orono, ME, January, 2014.

Letcher, B.H. et al. Brook trout population dynamics: Integrated modeling across scales and data types, **Invited**, Atlantic salmon Ecosystem research forum, Orono, ME, Jan 2014.

Letcher, B.H., Response of stream salmonids to environmental variation: patterns, sensitivities, and forecasts, **Invited keynote**, 16th annual meeting of the Centre Interuniversel du recherche au saumon atlantique. Quebec City, Quebec, May 2013

Letcher, B.H. et al., Monitoring Genetically Identifiable Atlantic Salmon in the Connecticut River, **Invited**, Biennial Atlantic salmon research forum, Hadley MA, February 2013

Letcher, B.H. et al., A modular Bayesian approach to estimating species response to environmental change, **Invited**, Annual meeting of the International Association of Landscape Ecologists, Newport, RI, April 2012

Letcher, B.H. et al., Modeling population persistence on the streamscape, **Invited**, USFWS Region 5 Seminar Series, Hadley, MA, April 2012

Letcher, B.H., Connectivity in stream networks, **Invited plenary** at the Quebec Society of Animal Behavior annual meeting, Sherbrooke Quebec, Nov 2011.

Letcher, B.H. et al., The Differential Importance of Stream Units and Their Connectivity for Population Persistence, within and trans-generation connectivity, **Invited**, Riverscapes Symposium American fisheries Society Annual Meeting, Seattle WA, Sept 2011.

Letcher, B.H., Effects of stream temperature and flow on stream salmon population persistence: can we forecast climate change effects? **Invited**, MA Rivers Alliance meeting – River monitoring and climate change in MA, Leominster, MA, May 2011.

Letcher, B. H., Brook trout population persistence despite isolation. **Invited**, University of Laval, Quebec, Canada, March 2011.

Letcher, B.H., et al. Variation in freshwater Atlantic salmon survival among rivers: timing, body size and extent, **Invited**, Biennial Connecticut River Research Forum, Hadley, MA Feb 2011.

Letcher, B. H., K.H. Nislow, Brook trout population persistence, how does it work? **Invited**, Annual meeting of the Eastern Brook Trout Joint Venture, Mountain Lake, VA, Oct 2010.

Letcher, B. H., J. Coombs, K.H. Nislow, Maintenance of phenotypic variability: size selection and quantitative genetics of stream-dwelling brook trout, Advances in the population ecology of stream salmonids, Lueca, Spain, May 2010.

Letcher, B. H., Population persistence in brook trout: models and forecasts. **Invited** Engineering Department Departmental Seminar, University of Massachusetts, Jan. 2010, Amherst, MA.

Letcher, B. H., Population persistence in brook trout: models and forecasts. **Invited** School of Biology and Ecology Departmental Seminar, University of Maine, Jan. 2010, Orono, ME.

Letcher, B. H., Population persistence in brook trout: models and forecasts. **Invited** Ecology and evolutionary Biology Departmental Seminar, University of Connecticut, Dec. 2009, Storrs, CT.

Letcher, B. H., J. Coombs, P. Schueller, K. Griswold, Monitoring Genetically Identifiable Atlantic Salmon in the Connecticut River, **Invited** presentation for Connecticut River Atlantic salmon Assessment Committee Meeting, Dec 2009, Turners Falls, MA

Letcher, B. H., K. Nislow, Seasonal variation in the relative importance of factors influencing growth, survival and movement of stream salmonids: towards a hierarchical model of population persistence. **Invited** Monster Seminar Jam, NOAA Northwest Science Center, October 2009, Seattle, WA.

Letcher, B. H., J. Coombs, P. Schueller, K. Griswold, Monitoring Genetically Identifiable Atlantic Salmon in the Connecticut River, Annual Atlantic salmon forum, Feb 2009, Hadley, MA

Letcher, B. H., K. Nislow, J. Coombs, T. D. Dubreuil, M. O'Donnell. Habitat-dependent growth and survival of brook trout and Atlantic salmon in a small stream – everybody in the pool? American Fisheries Society, 2008 Annual Meeting, Ottawa, ONT.

Letcher, B.H., Extinction risk due to stream fragmentation in a brook trout population, **Invited** presentation at the DNR/TNC seminar series, January 2008, Boston, MA

Letcher, B.H., K. Nislow, J. Coombs, M. O'Donnell, and T. Dubreuil. Habitat effects on resident brook trout persistence. Annual Atlantic salmon forum, January 2008, Orono, ME.

Letcher, B.H. and Nislow, K. Extinction risk due to stream fragmentation in a brook trout population. **Invited** talk, Trout Unlimited Annual Meeting, April 2007, Leominster, MA.

Letcher, B.H. Extinction risk due to stream fragmentation in a brook trout population. **Invited** Departmental Seminar, North Carolina State University, Feb. 2007, Raleigh, NC.

Letcher, B. H., K. Griswold, J. Coombs, Monitoring Genetically Identifiable Atlantic Salmon in the Connecticut River, Biennial Atlantic salmon forum, Feb 2007, Hadley, MA

Letcher, B. H., T. D. Dubreuil, M. O'Donnell, J. Zydlewski, B. Connery. Brook trout growth and movements in a Maine coastal stream, Stanley Brook. **Invited** talk, USFWS sea-run trout workshop, Dec. 2006, Hadley, MA.

Letcher, B. H., T. D. Dubreuil, M. O'Donnell, K. Nislow. Population consequences of Brook trout movements in a fragmented landscape. **Invited** talk, American Fisheries Society Meeting, Sept. 2006, Lake Placid, NY

Letcher, B.H, Mechanisms maintaining body size variation in stream salmonids. **Invited** talk, International Congress on the Biology of Fishes, July 2006, St. John's Newfoundland, CA

Letcher, B.H, Causes and consequences of body size variation in stream fish. **Invited** seminar, Organismic and Evolutionary Biology Program, University of Massachusetts, April, 2006.

Letcher, B. H., T. D. Dubreuil, M. O'Donnell, K. Nislow, Linking stream flow to fish population response: relative effects of flow and temperature variation on Atlantic salmon growth and survival in a small stream. 2005 WRRC conference, Amherst MA.

Letcher, B. H., Coombs, J., T. D. Dubreuil, M. O'Donnell, K. Nislow. Piecing together a salmonid production model for a small stream. **Invited** seminar, Trout unlimited Northeast region Annual Science meeting, 2005, Litchfield CT.

Letcher, B. H., Coombs, J., T. D. Dubreuil, M. O'Donnell, K. Nislow. Piecing together a salmonid production model for a small stream. **Invited** seminar, Harvard Forest seminar series, 2005, Petersham, MA.

Letcher, B. H., G. Horton, T. D. Dubreuil, M. O'Donnell, Selection on body size in juvenile Atlantic salmon: power of multi-state mark-recapture models to detect size dependent survival, Ecological Society of America, 2004 annual meeting, Portland Oregon.

Letcher, B. H., G. Horton, T. D. Dubreuil, M. O'Donnell, No selection on size or growth in juvenile Atlantic salmon after accounting for emigration and probability of capture using a multistrata mark-recapture model, Evolutionary and Ecological Ethology of Fishes 2004 meeting, Saudakraukur, Iceland

Letcher, B. H., T. D. Dubreuil, M. O'Donnell, M. Obedzinski, and K Nislow, Life History Consequences of Variation in Stocking Date for Stream-Dwelling Atlantic Salmon Parr, American Fisheries Society, 2003 Annual Meeting, Quebec City, Quebec, CAN.

Letcher, B.H, Sources of size variation in stream-dwelling Atlantic salmon parr. **Invited** seminar, Norway Institute of Nature Research, Trondheim, Norway, 2003

Letcher, B.H. Sources of size variation in stream-dwelling Atlantic salmon. **Invited** seminar, Department of Biology , University of Maine, May 2, 2003.

Letcher, B.H. Determinants of life history variation in stream-dwelling Atlantic salmon. **Invited** seminar for Biology Department, Worcester Polytechnic Institute, February 18, 2003

Letcher, B.H., T.D. Dubreuil, M.O. O'Donnell, M. Obedzinski and K. Nislow. Effects of timing of stocking on Atlantic salmon in the West Brook, MA. USFWS Forum on Atlantic Salmon research, Hadley, MA, February, 2003.

- Letcher, B.H. and Nislow, K.H. Determinants of growth and survival across scales: use of parentage assignment to assess heritable variation in juvenile Atlantic salmon. **Invited** presentation at SALGEN quantitative/experimental workshop, Arrendono, Spain, Sept. 2002.
- Letcher, B.H. Family differences in growth, survival and movement of stream-dwelling Atlantic salmon parr. EEEF meeting, Quebec City, Quebec, September, 2002.
- Letcher, B.H. T.D. Dubreuil, and G. Gries. Incorporation of PIT tag antennas into a long-term, detailed tagging study in a small stream. NMFS PIT tag antenna workshop, Manchester, WA, December, 2001.
- Letcher, B.H. and G. Gries. Life-history dependent morphometric variation in stream-dwelling Atlantic salmon. AIC meeting, Center Harbor, NH, September, 2001.
- Letcher, B.H. and G. Gries. Family differences in growth and survival of individually-tagged juvenile Atlantic salmon. USFWS Forum on Salmon research, January, 2001.
- Letcher, B.H. and G. Gries. Sources of growth variation in individual Atlantic salmon parr in West Brook, MA, USA, International Congress on the Biology of Fishes, Aberdeen, Scotland, 2000.
- Letcher, B.H., F. Juanes and G. Gries. Growth, survival and maturity of individual Atlantic salmon parr in West Brook, MA, USA. Canadian Conference for Fisheries Research, Fredericton, New Brunswick, 2000.
- Letcher, B.H., and T.L. King. Targeted stock identification using multilocus genotype 'familyprinting'. AFS meeting, Hartford, CT, 1998.
- Letcher, B.H., G. Gries, and F. Juanes. Parr maturation at low latitudes: growth, smolting and age-0 maturation. AFS meeting, Hartford, CT, 1998.
- Letcher, B.H., G. Gries, T.D Terrick, and F. Juanes. Growth movement and maturity of individual Atlantic salmon parr in West Brook, MA, USA. International congress on the biology of fish, Baltimore, MD, 1998.
- Letcher, B.H., and T. King. Tools for stream fish restoration: Atlantic salmon genetics, broodstock management and fry marking. Ecology of Stream fish, state of the art and future prospects, 1998, Luarca, Spain.
- Juanes, F., and Letcher, B.H. Ecology of stream fish: insights gained form an individual-based approach. Ecology of Stream fish, state of the art and future prospects, 1998, Luarca, Spain.
- Letcher, B.H, T.L King. Tools for evaluation of anadromous fish restoration: genetics, broodstock management and fry marking. **Invited** seminar at Dartmouth College, Fall 1997, and at the University of Rhode Island, Winter 1998.

Letcher, B.H, T.L King. Tools for evaluation of anadromous fish restoration: potential for large-scale genetic marking of Connecticut River Atlantic Salmon. ICES Science Meeting, 1997 (nominated for outstanding paper).

Letcher, B.H, T.L King. Genetic variability of Connecticut River Atlantic salmon and potential for a genetic mark to identify stocking tributary. American Fisheries Society Meeting, 1997.

Letcher, B.H, T. Terrick. Effects of fry developmental stage on stocking success, American Society of Ichthyologists and Herpetologists meeting, 1997.

Letcher, B.H, J. A. Rice, L.B. Crowder, and F.P. Binkowski. Size-dependent effects of continuous and intermittent feeding on starvation time and mass loss in starving yellow perch (*Perca flavescens*) larvae and juveniles. American Society of Ichthyologists and Herpetologists meeting, 1996.